PROJECT PROPOSAL

Enterprise QR Engagement & Rewards Platform

Dynamic QR campaigns for customer engagement, loyalty, feedback, giveaways and digital reward fulfillment

Date: September 29, 2026

COMMERCIAL PRICE AND DELIVERY TIMELINE: #70,000 4-6 weeks

As a result of building another qr workflow from the ground up and another app itself

To be paid after completion

Concept Summary

A business-grade platform where an administrator creates a unique QR campaign for a business, organization, individual, leader, branch, event or initiative. When a person scans the QR code, they see a branded information page, choose an action such as Claim Reward, Give Feedback, Join Giveaway or Complete Survey, submit the required information, and - if eligible - receive a configured benefit such as airtime, mobile data, a utility credit, coupon, voucher, gift card or loyalty reward.

Prepared as an expanded continuation for the original QR Code Generator & Management SaaS proposal.

1. Project Overview

The original proposal described a QR generation and management SaaS with user accounts, dynamic QR redirects and scan analytics. This continued scope expands that foundation into a multi-tenant engagement and reward platform built around configurable QR workflows.

The system is not merely a QR image generator. Each QR code becomes the entry point into a managed campaign with its own branding, information page, forms, eligibility rules, reward configuration, budget, analytics and lifecycle.

Primary objectives

- Allow administrators to create and manage QR campaigns for many different clients, organizations, branches, people or initiatives from one platform.
- Give every QR campaign a branded public landing page containing the relevant business or campaign information.
- Support configurable actions such as rewards, loyalty claims, complaints, feedback, surveys, giveaways, lead capture and promotional campaigns.
- Collect campaign-specific participant information with consent and validation.
- Automatically fulfill eligible rewards through integrated third-party providers and record the result.
- Prevent duplicate claims, automated abuse and uncontrolled reward spending.
- Provide campaign analytics, submissions, reward history, exports and audit logs.
- Keep the architecture flexible enough to support future SaaS, white-label, API and enterprise capabilities.

2. Users and Roles

Role	Purpose	Typical capabilities
Platform Super Admin	Operates the entire system.	Manage organizations, admins, provider integrations, limits, global settings, audits and platform analytics.

Organization / Client Admin	Manages campaigns for a business, organization or initiative.	Create QR campaigns, edit landing pages, configure forms/rewards, fund or assign budgets, view analytics and export submissions.
Campaign Manager / Staff	Optional restricted team member.	Manage assigned campaigns, review complaints/submissions and monitor fulfillment according to permissions.
Participant / Customer	Scans a QR code.	View public information, complete an allowed workflow and claim a reward if eligible.

3. Core Campaign Model

An administrator can create many independent QR campaigns. A campaign can belong to a business, organization, person, branch, event or other client entity. The same organization may have several QR codes with different purposes, rules and rewards.

Campaign setting	Examples
Identity	Campaign name, organization, branch, owner, internal reference.
Public branding	Logo, cover image, colors, description, address, website, social links, contact details.
QR configuration	Dynamic short URL, QR design, downloadable PNG/SVG/PDF asset, active/inactive state.
Workflow	Reward claim, loyalty reward, feedback, complaint, giveaway, survey, lead capture, event promotion.
Form schema	Name, age/age range, occupation, phone, email, location, customer ID, rating, complaint text, custom questions.
Eligibility	Start/end date, geography if appropriate, customer group, unique phone, invite code, transaction/reference requirement.
Reward	Airtime, data, utility credit/payment, coupon, voucher, gift card, discount, points or no monetary reward.
Controls	Reward value, maximum claims, per-person limit, campaign budget, daily limit, verification requirement.
Status	Draft, scheduled, active, paused, completed, archived.

Example use cases

- Loyalty: a restaurant places a QR code at checkout; returning customers submit their phone number and receive a configured airtime or coupon reward.
- Complaint and service recovery: a customer scans a branch-specific QR, submits a complaint and may receive a goodwill voucher after review.

- Giveaway: a campaign organizer distributes a QR code; eligible participants verify their phone number and receive a limited reward while stock/budget lasts.
- Customer research: a business collects age range, occupation and preference data through a survey, optionally rewarding completed submissions.
- Community distribution: an organization or public-facing leader uses a controlled QR campaign to distribute airtime/data benefits with eligibility and anti-duplicate rules.
- Events: each event or booth has a QR page containing event information, feedback and sponsor rewards.

4. End-to-End Participant Experience

Step	Participant experience	System action
1. Scan	Customer scans the printed or digital QR code.	Dynamic route resolves the campaign and records a privacy-conscious scan event.
2. Landing page	Branded page shows business/campaign information.	System loads only active public campaign content.
3. Choose action	User taps Rewards, Feedback, Complaint, Survey, Giveaway or another enabled action.	Workflow-specific rules and form schema are loaded.
4. Submit details	User provides required information and consent.	Backend validates fields, normalizes phone data and applies rate limits.
5. Verify	If required, user completes OTP or another verification step.	System checks duplicate identity/phone/device indicators and campaign eligibility.
6. Claim	Eligible participant confirms reward claim.	Backend reserves campaign budget and creates an idempotent reward transaction.
7. Fulfill	Participant receives airtime/data/voucher/etc.	Reward provider is called; response/reference is stored. Failed eligible transactions can be retried safely.
8. Confirmation	Success, pending or failure status is shown.	Analytics and audit history are updated without double-paying the same claim.

Important payment architecture

Paystack can be used for collecting money or funding flows where appropriate, but the implementation should not assume that Paystack itself is the airtime/data fulfillment engine. Airtime, data and many utility products normally require a VTU/bill-payment/reward API or another provider that explicitly supports those products. The final provider should be selected based on supported Nigerian networks/services, API reliability, settlement model, pricing, webhooks, transaction lookup, reversals and production credentials.

5. Reward Engine

- Reward catalog: airtime, mobile data, utility credit/payment, coupon code, voucher, gift card, discount, loyalty points or custom benefit.
- Fixed or campaign-defined value, for example a ₦100 airtime reward.
- Network/provider selection can be automatic from verified phone data where supported, or explicitly selected by the participant.
- Campaign-level budget, maximum number of rewards, daily cap and optional per-reward inventory.
- Claim states such as initiated, verification_pending, eligible, reserved, processing, successful, failed, reversed, rejected and expired.
- Idempotency keys so a browser refresh or network retry cannot create duplicate payouts.
- Provider transaction reference, request/response status, retry count and reconciliation status stored for every fulfillment.
- Optional manual approval for high-value or complaint-resolution rewards.

6. Configurable Data Collection

Forms should be campaign-driven rather than hard-coded. The admin chooses which fields are shown, which are required and which workflow they belong to.

Field category	Examples
Identity	Full name, age or age range, gender only where legitimately needed, occupation.
Contact	Phone number, email address.
Location	State, LGA, city or branch; precise address only where genuinely required.
Customer context	Customer ID, receipt/reference number, membership number, product/service used.
Feedback	Rating, category, complaint, suggestion, free-text comments.
Survey	Single choice, multiple choice, yes/no, short text, long text and custom questions.
Consent	Privacy notice acceptance, reward terms, optional marketing consent kept separate from required processing consent.

Data minimization is important: campaigns should collect only information needed for their stated purpose. Age range is often preferable to exact date of birth where exact age is unnecessary.

7. Abuse Prevention and Eligibility Controls

- Phone OTP verification where the risk/value justifies it.
- One reward per verified phone number, customer reference or configured period.

- Rate limiting by IP/session/device signals without treating device fingerprinting as infallible identity.
- CAPTCHA or bot challenge for suspicious/high-volume traffic.
- Campaign invite codes, receipt/reference validation or pre-approved customer lists where needed.
- Server-side eligibility checks; never trust reward eligibility calculated only in the browser.
- Atomic budget reservation before fulfillment to prevent race conditions and overspending.
- Velocity rules for repeated claims, unusual patterns and multiple failed verification attempts.
- Admin review queue for suspicious or high-value claims.
- Immutable audit trail for material admin changes and reward decisions.

8. Administration Dashboard

Module	Key functions
Organizations	Create/manage clients, profiles, branding, admins and status.
Campaigns	Create, duplicate, schedule, activate, pause, archive and search campaigns.
QR Codes	Generate/download assets, change campaign destination/content without reprinting, deactivate compromised codes.
Landing Pages	Edit business details, hero content, contact links, workflow buttons and terms.
Form Builder	Add/reorder fields, required rules, validation and consent text.
Rewards	Configure reward type/value/provider, budget, inventory, caps and approval rules.
Submissions	Filter, search, review and export participant responses.
Fulfillment	View reward transactions, failures, retries, provider references and reconciliation.
Analytics	Scans, unique participants, conversion, submissions, successful claims, reward spend and campaign performance.
Audit & Security	Admin activity, role permissions, suspicious claims and security events.

9. Proposed Technical Architecture

The existing Next.js/React direction can be retained, while the backend is expanded into campaign, form, eligibility and reward services.

Layer	Recommended approach	Purpose

Frontend	Next.js + React + TypeScript	Admin dashboard and responsive public campaign pages.
UI	Tailwind CSS + component library	Reusable enterprise dashboard and mobile-first scan experience.
Backend	Next.js server/API routes or Node.js service	Authentication, campaign APIs, form submission, eligibility and reward orchestration.
Database	MongoDB Atlas	Organizations, campaigns, forms, scans, submissions, claims, transactions and audit logs.
Authentication	Secure sessions/JWT + RBAC	Admin authentication, organization isolation and permission enforcement.
QR routing	Dynamic short route, e.g. /q/{code}	Stable QR destination that can resolve campaign state without reprinting.
Verification	SMS/OTP provider where enabled	Phone ownership verification and anti-abuse support.
Rewards	Provider adapter layer	Connect one or more VTU, bill-payment, coupon or gift-card providers without coupling the app to one vendor.
Payments	Paystack where applicable	Funding, checkout or other supported payment flows; separate from reward fulfillment unless the chosen product is explicitly supported.
Layer	**Recommended approach**	**Purpose**
Jobs/Queue	Background job/queue service	Reliable fulfillment, retries, webhook processing and reconciliation.
Deployment	Vercel + managed services	Production hosting, logs, secrets and scalable application delivery.

10. Core Data Model

Entity	Important data
Organization	name, branding, contacts, status, owner, settings
AdminUser	organizationId, role, permissions, authentication/security fields
Campaign	organizationId, slug/code, workflow type, content, dates, status, eligibility, limits

FormDefinition	campaignId, fields, validation rules, consent/terms versions
ScanEvent	campaignId, timestamp, coarse technical metadata where permitted
Submission	campaignId, participant fields, consent version, verification status, review state
RewardConfig	campaignId, type, value, provider, total budget, daily cap, inventory
Claim	submissionId, eligibility result, deduplication keys, state, reserved amount
RewardTransaction	claimId, provider, idempotency key, provider reference, amount, status, retries
AuditLog	actor, organization, action, target, timestamp, security-relevant context

Recommended product framing

Position the system as an Enterprise QR Engagement & Rewards Platform, not simply a QR generator. The QR code is the access channel; the real product is the configurable campaign, data-collection, eligibility, fulfillment and analytics engine.

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